
Linear Statistical Models

B. Statistical Data Science 2nd Year Indian Statistical Institute

Teacher: Soham Sarkar

Exercise Series 6

The following is a revision of what was done in the class.

Exercise 1. Consider a linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon} \sim (\mathbf{0}, \sigma^2 \mathbf{I})$ and $\mathbf{X}$ is not of full column rank. We have seen three strategies to deal with such non-full-rank cases.

- **Strategy 1:** Use generalized inverse to get a solution to the normal equations:

$$\mathbf{X}^\top \mathbf{X} \boldsymbol{\theta} = \mathbf{X}^\top \mathbf{y} \Rightarrow \hat{\boldsymbol{\theta}} = (\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top \mathbf{y}.$$

- **Strategy 2:** Reparameterize to get a full-rank model: $(\mathbf{y}, \mathbf{X}\boldsymbol{\theta}, \sigma^2 \mathbf{I}) \Rightarrow (\mathbf{y}, \mathbf{Z}\boldsymbol{\gamma}, \sigma^2 \mathbf{I})$, where $\mathbf{Z}$ is full column rank and $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{Z})$. In this case, $\boldsymbol{\gamma}$ is a linear function of $\boldsymbol{\theta}$. Since $\mathbf{Z}$ is of full column rank, $\boldsymbol{\gamma}$ can be uniquely estimated from the reparameterized model.
- **Strategy 3:** Impose restriction $\mathbf{H}\boldsymbol{\theta} = \mathbf{0}$, where the rows of $\mathbf{H}$ are linearly independent, and are linearly independent with the rows of $\mathbf{X}$ (or, equivalently, with the rows of $\mathbf{X}^\top \mathbf{X}$). This ensures that the parameters are uniquely determined.

All these three strategies lead to the same set of estimators for the elementary contrasts $\tau_i - \tau_j$ in the one-way model $y_{ij} = \mu + \tau_i + \epsilon_{ij}, i = 1, \dots, I, j = 1, \dots, n_i$. This in turn shows that the estimator of any contrast is the same for the three strategies. Here, we generalize this result to all estimable linear parametric forms $\mathbf{a}^\top \boldsymbol{\theta}$.

- (a) Show that the estimator $\mathbf{a}^\top \hat{\boldsymbol{\theta}}_{(1)}$ for any estimable linear parameter $\mathbf{a}^\top \boldsymbol{\theta}$ via Strategy 1 does not depend on the choice of the generalized inverse $(\mathbf{X}^\top \mathbf{X})^-$.

Hint: Recall the condition on $\mathbf{a}$ for $\mathbf{a}^\top \boldsymbol{\theta}$ to be estimable.

- (b) Show that it is possible to obtain an estimator of any estimable linear parametric form $\mathbf{a}^\top \boldsymbol{\theta}$ via Strategy 2, and the estimator does not depend on the choice of reparameterization.

Hint: Use the relationship between $\mathbf{X}$ and $\mathbf{Z}$. What is $\mathbf{a}^\top \boldsymbol{\theta}$ in terms of $\boldsymbol{\gamma}$?

- (c) Show that the estimators of $\mathbf{a}^\top \boldsymbol{\theta}$ obtained in parts (a) and (b) are the same. Hence, for any estimable parameter, Strategy 1 and Strategy 2 lead to the same estimator.
- (d) Show that the estimator $\hat{\boldsymbol{\theta}}_{(3)}$ via Strategy 3 is the unique vector that satisfies both $\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\theta}}_{(3)} = \mathbf{X}^\top \mathbf{y}$ and $\mathbf{H} \hat{\boldsymbol{\theta}}_{(3)} = \mathbf{0}$. Use this to show that the estimator $\mathbf{a}^\top \hat{\boldsymbol{\theta}}_{(3)}$ of any estimable linear parameter $\mathbf{a}^\top \boldsymbol{\theta}$ does not depend on the choice of the restriction imposed.
- (e) Part (d) shows that the estimator obtained via Strategy 3 is similar to that of Strategy 1, except we impose some restrictions to uniquely determine the generalized inverse. This, together with part (c), shows that the estimator of any estimable linear parametric form using all the three strategies coincide.
- (f) Consider the problem of testing the hypothesis $\mathcal{H}_0 : \mathbf{A}\boldsymbol{\theta} = \mathbf{0}$ against $\mathcal{H}_a : \mathbf{A}\boldsymbol{\theta} \neq \mathbf{0}$. We say that the hypothesis is *testable* if $\mathbf{A}\boldsymbol{\theta}$ is estimable (i.e., all the components of $\mathbf{A}\boldsymbol{\theta}$ are estimable). Verify that the following strategy remains valid for obtaining the F -test statistic for testing $\mathcal{H}_0$ vs. $\mathcal{H}_a$.

- Fit the full model $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon}$.
 - Calculate the sum of squared errors: $\text{SSE} = \min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|^2$.
 - Calculate the degrees of freedom: $\text{df} = n - k$, where $k = \text{rank}(\mathbf{X})$.
- Fit the restricted model under $\mathcal{H}_0$.
 - Calculate the sum of squared errors: $\text{SSE}_{\mathcal{H}_0} = \min_{\boldsymbol{\theta}: \mathbf{A}\boldsymbol{\theta}=\mathbf{0}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|^2$.
 - Calculate the degrees of freedom: $\text{df}_{\mathcal{H}_0} = n - (k - q)$, where q is the number of (linearly independent) restrictions imposed by $\mathcal{H}_0$, which is same as the (row-)rank of $\mathbf{A}$.
- Compute $\text{SS}(\mathcal{H}_0) = \text{SSE}_{\mathcal{H}_0} - \text{SSE}$, $\text{df}(\mathcal{H}_0) = \text{df}_{\mathcal{H}_0} - \text{df}$.
- Define $T = \frac{\text{SS}(\mathcal{H}_0)/\text{df}(\mathcal{H}_0)}{\text{SSE}/\text{df}} = \frac{\text{SS}(\mathcal{H}_0)/q}{\text{SSE}/(n - k)}$. T is the F -test statistic.

Exercise 2. In this exercise, we study the Gauss-Markov theorem under the general setup $(\mathbf{y}, \mathbf{X}\boldsymbol{\theta}, \sigma^2\mathbf{I})$ where $\mathbf{X}$ is not necessarily of full column rank. Recall that a linear parametric form $\mathbf{a}^\top \boldsymbol{\theta}$ under the setup $(\mathbf{y}, \mathbf{X}\boldsymbol{\theta}, \sigma^2\mathbf{I})$ is estimable if and only if $\mathbf{a} \in \mathcal{R}(\mathbf{X})$. We have also seen in the class that for an estimable parametric form $\mathbf{a}^\top \boldsymbol{\theta}$, the least squares estimator $\mathbf{a}^\top \hat{\boldsymbol{\theta}}_{\text{LS}}$ is an linear and unbiased estimator. Here, $\hat{\boldsymbol{\theta}}_{\text{LS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ is a least-squares estimator, which is one (out of possibly infinitely many) of the solutions to the normal equations $\mathbf{X}^\top \mathbf{X}\boldsymbol{\theta} = \mathbf{X}^\top \mathbf{y}$. And, although $\hat{\boldsymbol{\theta}}_{\text{LS}}$ may not be unique, $\mathbf{a}^\top \hat{\boldsymbol{\theta}}_{\text{LS}}$ is always unique. We have also derived the variance of the estimator $\mathbf{a}^\top \hat{\boldsymbol{\theta}}_{\text{LS}}$. Show that $\mathbf{a}^\top \hat{\boldsymbol{\theta}}_{\text{LS}}$ is the best linear unbiased estimator (BLUE) of $\mathbf{a}^\top \boldsymbol{\theta}$. That is, it has the least variance among all linear unbiased estimators of $\mathbf{a}^\top \boldsymbol{\theta}$.

Hint: Try to mimic the proof for the full column rank model.

Extend this result to the case where the linear parameter is a vector. Consider an estimable parametric form $\mathbf{A}\boldsymbol{\theta}$. Show that the least-squares estimator $\mathbf{A}\hat{\boldsymbol{\theta}}_{\text{LS}}$ is the best linear unbiased estimator (BLUE) of $\mathbf{A}\boldsymbol{\theta}$. That is, for every linear unbiased estimator $\tilde{\boldsymbol{\eta}}$ of $\mathbf{A}\boldsymbol{\theta}$, $\text{Var}(\tilde{\boldsymbol{\eta}}) \succeq \text{Var}(\mathbf{A}\hat{\boldsymbol{\theta}}_{\text{LS}})$.

Exercise 3. A researcher wants to study the effect of five different diets on the body fat percentage. For this study, she gathers some volunteers from different age groups and assigns different diets to each of them. The numbers of volunteers allocated to the different diets are shown in the table below.

Age group	Diet 1	Diet 2	Diet 3	Diet 4	Diet 5	Total
10–15	0	0	1	1	1	3
16–20	1	1	1	1	0	4
21–25	2	2	1	2	2	9
26–30	3	2	2	2	3	12
31–35	1	2	2	1	1	7
36–40	1	1	0	1	1	4
Total	8	8	7	8	8	39

After 3 months of following the diet, the researcher notes the change in body fat percentage of the volunteers and uses the model

$$y = \mu + \tau_i + \delta_j + e \quad \text{if the volunteer receives Diet } i \text{ and is in the } j\text{-th age-group.}$$

Here, y is the change in body fat percentage and e is the error term.

- (a) Write the model in vector-matrix notation. Clearly define the parameter vector $\boldsymbol{\theta}$ and the design matrix $\mathbf{X}$.

- (b) Formulate the normal equations. Is it possible to solve the normal equations? Are all the parameters of the model estimable?
- (c) Are the elementary contrasts involving τ_i 's estimable? If so, then find the best linear unbiased estimators of these elementary contrasts.

Hint: Recall that the elementary contrasts involving the τ_i 's are of the form $\tau_i - \tau_j$ for $i, j = 1, \dots, 5$. All the elementary contrasts are estimable if and only if $\tau_1 - \tau_j$ is estimable for all $j = 2, \dots, 5$.

- (d) Is it possible to test the hypothesis that all the diets have the same effect on body fat percentage, i.e., $\mathcal{H}_0 : \tau_1 = \dots = \tau_5$ vs. $\mathcal{H}_a : \text{not } \mathcal{H}_0$, under the formulated model? If so, then find the F -test statistic for testing $\mathcal{H}_0$ against $\mathcal{H}_a$. Find its distribution under the assumption of normality of the error e . Determine an appropriate cut-off which ensures that the test is of level $\alpha \in (0, 1)$.

Note: Here, the age-groups are used as *blocks* to reduce heterogeneity. In practice, it is not always possible to ensure that all the *treatments* can be applied to all the *blocks*, or vice-versa. This may be due to unavailability of experimental units, cost/time constraints etc. When some of the block-treatment combinations are missing, such an experimental design is known as an *incomplete block design*.